

Human Body Systems: Breathing & Transport

Quick reference sheet · Biology · keep handy for revision

1. Respiration

Aerobic (with oxygen): glucose + oxygen → carbon dioxide + water (+ *energy released*)

Anaerobic (no oxygen, e.g. sprinting): glucose → lactic acid (+ *energy released, but much less*)

Anaerobic respiration is fast to start but releases far less energy, and the lactic acid that builds up causes muscle fatigue/cramp.

2. Exchange surfaces & the alveoli

Every good exchange surface has: a **large surface area**, **thin walls** (short diffusion distance), a **good blood supply** (keeps the concentration gradient steep), and a **moist lining** (lets gases dissolve).

The alveoli in the lungs show all four: millions of them give a huge surface area; walls one cell thick give a short diffusion path; a network of capillaries gives a good blood supply; the lining is moist.

3. The heart & double circulation

Chamber	Receives / pumps to
Right atrium	Receives deoxygenated blood from the body
Right ventricle	Pumps deoxygenated blood to the lungs (pulmonary circuit)
Left atrium	Receives oxygenated blood from the lungs
Left ventricle	Pumps oxygenated blood to the body (systemic circuit) — thicker, more muscular wall than the right ventricle, since it must reach the whole body

Double circulation: blood passes through the heart twice on each full circuit of the body — once via the lungs, once via the rest of the body.

4. Blood vessels

Vessel	Wall & lumen	Job
Artery	Thick, muscular & elastic wall; narrow lumen	Carries blood at high pressure away from the heart
Vein	Thin wall; wide lumen; has valves	Carries blood at low pressure back to the heart; valves stop backflow
Capillary	Wall just one cell thick	Links arteries to veins; substances diffuse into/out of nearby cells

5. Blood components

Component	Looks like	Function
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Red blood cell	Biconcave disc, no nucleus, full of haemoglobin	Carries oxygen around the body
White blood cell	Larger, has a nucleus	Fights infection (engulfs pathogens / makes antibodies)
Platelet	Small cell fragment, no nucleus	Helps blood clot at a wound
Plasma	Pale yellow liquid	Transports blood cells, dissolved food, CO ₂ , urea & hormones